Mohamed Sucule

Frontend Engineer · React · TypeScript · Accessibility · Performance

Birmingham, UK SC Security Cleared UK Right to Work mohamedsucule@gmail.com
linkedin.com/in/mohamed-sucule-bb8747439 github.com/Mohamed-Sucule mohamed-sucule.github.io/Projects

PROFILE

Frontend engineer with 4+ years' commercial experience across product and consultancy teams, including SC-cleared public-sector programmes at Capgemini. I build in React and TypeScript, make interfaces that work from the keyboard and pass accessibility checks, and profile before I optimise: in my own apps that took a spreadsheet recalculation from 681 ms to 133 ms and a map render from 41.7 s to 0.39 s.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript (ES2022+), HTML, CSS (Grid, Flexbox, custom properties), Python
Frontend	React 19 (hooks, useReducer, useSyncExternalStore), Vite, npm workspaces, mobile-first layout
Web platform	Canvas 2D, SVG, WebGL, Web Workers, OffscreenCanvas, IndexedDB, Web Audio
Accessibility	WCAG 2.1 AA, WAI-ARIA patterns (grid, combobox, tabs), keyboard navigation, axe-core in CI
Performance	Chrome DevTools profiling, virtualised lists and grids, typed arrays, off-main-thread rendering
Testing & CI	Vitest, Playwright, fast-check, ESLint, Prettier, GitHub Actions, GitLab CI/CD
Cloud & delivery	REST APIs, AWS, Azure, Git, Agile delivery, technical discovery, stakeholder management

EXPERIENCE

Software Engineer · Capgemini — July 2022 – Present

- Build and maintain production web applications and supporting services in TypeScript, JavaScript and Python, with a focus on clear, reliable user-facing functionality.
- Connect front-end experiences to production back ends through REST APIs, cloud infrastructure and CI/CD pipelines.
- Turn requirements into technical solutions with senior stakeholders, from discovery to deployment and live support.
- Deliver for major public- and private-sector clients in Agile teams, including programmes under SC security clearance.

Web Developer · SynchrTrade — 2021 – 2022

- Built responsive front-end features in React and JavaScript, working closely with back-end engineers and designers.
- Refined user-facing features in agile sprints, balancing usability, technical constraints and delivery speed.

SELECTED PROJECTS

Cells · a spreadsheet written from scratch · React, TypeScript, Vitest, Playwright — Live

- Formula engine from scratch: a Pratt parser, 116 Excel functions and a graph that recalculates only what changed.
- Virtualised ARIA grid of 1,048,576 rows with Excel's keyboard model, IndexedDB saving and live cross-tab editing.
- Profiled a 200,001-cell recalculation from 681 ms to 133 ms; 209 unit tests, including a property test against a reference evaluator, and 16 end-to-end tests with axe.

Your Place · print-ready map posters · React, TypeScript, Web Workers, OffscreenCanvas — Live

- Vector-tile decoder and renderer written from scratch, painting in a Web Worker so the map never waits on drawing.
- Profiling took the first poster from 41.7 s to 0.39 s; accessible combobox search in a shared component package.

Ember · an action game with no engine or asset files · Canvas 2D, Web Audio, Gamepad API — Live

- Real-time lighting from visibility polygons, generated maps and a live-synthesised score; keyboard, touch or gamepad.
- Tests play whole games with an autopilot and fail if any attack lands with less than 0.3 s of warning.

Portfolio · 37 projects that run in the browser — Live

- Tools, games and in-browser ML apps; 1,000+ tests and an accessibility scan of every page gate each deploy.

CERTIFICATIONS

AWS Certified Cloud Practitioner · Microsoft Azure AI Fundamentals (AI-900) · GitLab Certified

EDUCATION

BEng Mechanical Engineering · Aston University, Birmingham — 2021